Rowan Yu

Senior Unreal Engine Software Engineer | XR Developer Relations & Tooling | Performance

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PROFESSIONAL SUMMARY

Senior Unreal Engine Software Engineer with 8+ years of experience building VR/XR simulation systems, multiplayer architectures, developer-facing OpenXR plugins/samples, reusable Unreal tools and performance-sensitive real-time applications. At Big X Reality, developed military VR training simulators for mixed PC and PC-to-VR streamed workflows supporting up to 30 connected computers. Former HTC VIVE R&D engineer with hands-on developer relations experience: ViveOpenXR Unreal plugins, developer samples, tutorials and developer community support. Targeting Developer Relations Engineer roles that combine game development expertise, Android development and developer advocacy.

CORE SKILLS

Engines & languages: Unreal Engine 4/5, C/C++, Blueprint, C#, Unity

SDK/plugin architecture: ViveOpenXR, SRWorks Unreal Plugin, reusable Unreal plugins, developer samples, technical tutorials

Android development & validation: UE 5.4.4 VR Template, VIVE OpenXR 2.5.0, Android ASTC packaging, ADB, Logcat, Vulkan-only Android build, Android Device Profile override, OpenXR runtime troubleshooting; planned Google Pixel phone build/deploy and performance diagnostics (Perfetto)

Multiplayer: Replication, RPC, server/client launch workflows, cross-client synchronization, PC-to-VR streamed workflows

XR/VR: OpenXR, SteamVR, VIVE Pro 2, VIVE Focus 3, VIVE Tracker, SRWorks

Performance: Unreal Insights, profiling, async processing, multithreading, NPC optimization, runtime tuning

Tools & collaboration: packaged-build launch tooling, config/Excel tuning, test and validation workflows (Jenkins CI, Postman API testing, ClickUp/Trello bug and progress tracking); collaboration and developer advocacy: developer community support, mentoring, code/Blueprint review, cross-functional debugging, developer/customer validation support

Hardware/data integration: shared-memory IO, motion platform, Arduino, Antilatency, OptiTrack, HTTP, Socket.IO, TCP/UDP

Languages & multilingual communication: Mandarin Chinese (Native), English (Professional working proficiency)

ANDROID VALIDATION

- Independently validated a standalone Android VR workflow — UE 5.4.4 VR Template with VIVE OpenXR 2.5.0 on VIVE Focus 3 — through Android ASTC packaging, ADB install and Logcat crash diagnostics; resolved Vulkan/OpenXR runtime configuration, stereo rendering and Android Device Profile issues to reach a working baseline. Next planned track: Google Pixel phone build/deploy and performance diagnostics.

PROFESSIONAL EXPERIENCE

Big X Reality — Senior Unreal Engineer | 08/2022 – Present | New Taipei City, Taiwan

- Developed multiplayer VR training simulators for high-fidelity Windows PC and PC-to-VR streamed workflows, including VIVE Focus 3 environments, with performance-sensitive rendering and client/server synchronization.
- Developed large-scale multiplayer training systems supporting up to 30 connected computers across desktop and PC-VR clients; used Unreal Insights to trace and resolve performance and network transmission bottlenecks.
- Designed and implemented NPC behavior systems for 21 behavior trees across 11 training modes; built reusable NPC optimization/component workflows for large-scale VR training scenarios.
- Created a data collection abstraction layer that let teammates attach telemetry capture to gameplay features, then transformed, encrypted, timestamped and routed data through HTTP/Socket.IO pipelines to backend systems.
- Integrated Arduino-based weapon signals, Antilatency and OptiTrack tracking data into Unreal gameplay systems, resolving coordinate-space mismatches between external tracking systems and headset / VR runtime tracking spaces.
- Built packaged-build testing tools, including batch-script launchers for dedicated servers and selected client types, allowing teammates to quickly reproduce multiplayer scenarios without manually launching each executable.
- Built runtime configuration tools that allowed non-Unreal teammates to tune packaged EXE experiences through external Excel/config files during customer validation sessions, avoiding rebuilds and enabling faster feedback cycles.
- Mentored 2 junior engineers on Unreal development practices, Perforce-based Blueprint/binary asset workflows, Jenkins, Git setup and project workflows.

HTC Corporation — Engineer, VIVE R&D | 06/2019 – 08/2022 | New Taipei City, Taiwan

- Developed ViveOpenXR Plugin for UE4.27 and UE5.0 (OpenXR Scene Understanding, Facial Tracking, VIVE Cosmos controller mapping); submitted the UE5.0 plugin as a pull request to the EpicGames/UnrealEngine GitHub repository.
- Ported SRWorks Scene Understanding C++ low-level APIs into the OpenXR architecture, adapting API behavior and integration patterns to comply with OpenXR design rules.
- Implemented the OpenXR Facial Tracking Module using Unreal Live Link to stream facial expression data into Unreal Engine animation workflows.
- Created 4 developer-facing ViveOpenXR feature samples for UE4.27 and UE5.0 — Facial Tracking with Avatar/MetaHuman, Scene Understanding, Hand Tracking and Eye Gaze — to help Unreal developers adopt HTC-supported OpenXR features.
- Sole Unreal engineer on the team and the technical point of contact for HTC's Unreal developer community: answered questions and feature requests, debugged integration issues, and updated plugins, samples and tutorials based on developer feedback.
- Authored Unreal-focused ViveOpenXR tutorials across 4 feature areas (Facial Tracking, MetaHuman integration, Scene Understanding, Hand Tracking), reducing developer adoption friction for HTC-supported OpenXR features.

Artlord Studio — Engineer, R&D | 07/2016 – 12/2018 | Kaohsiung, Taiwan

- Developed 7 VR/Kinect games and interactive experiences — game development spanning gameplay, animation, VFX and multiplayer systems (UE4 Replication, VR IK, network profiling), deployed commercially at VIVELAND.
- Independently managed client communication, requirements and ~80% of game content for a Kinect climbing title; integrated HTC VIVE with Kinect, Fingo hand tracking and a motion platform.
- Evolved a Global Game Jam prototype into a publicly operated location-based VR experience integrated with a single-axis motion platform.

EDUCATION

Southern Taiwan University of Science and Technology — B.S. Multimedia and Entertainment Science